AMAZON ONLINE PROFITABILITY ANALYSIS REPORT

Group 10

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Table of Contents

Introduction 2

Data Exploration with Tableau 2

Statistical Analysis 2

Introduction to the Data 2

Descriptive Statistics 3

Attempted Models 3

Final Regression 3

Final Population Regression 3

Final Estimated Sample Regression 3

R^2 and Standard Error Interpretations 4

Significance Testing 4

Coefficient Interpretations 4

Common Violation Checks 5

Heteroskedasticity and Non-Linear Patterns 5

Endogeneity and Multicollinearity 5

Relevance of Findings 5

Important Trends and Relationships 5

Surprising Findings 5

Important Non-Significant Variables 6

Recommendations 6

Next Steps 6

Future Testing 6

Conclusion 6

Recap of Findings and Next Steps 6

Contact Information 7

Appendix 7

Sources Consulted 9

Introduction

This report analyzes a subset of historical Amazon order data from the United States, spanning the Office Supplies, Furniture, and Technology product categories. The dataset contains information from approximately 10,000 past customer orders and includes operational, geographic, and financial variables such as order date, shipping mode, customer segment, city, state, region, quantity ordered, revenue, discount percentage, and profit. The primary objective of this analysis was to develop a regression model capable of predicting average profit per item and to identify which factors most strongly influence whether an order is profitable. To achieve this, multiple regression models were tested using various combinations of explanatory variables. The final model gets just 26[1]% of the way towards perfectly predicting average profit per item, with shipping mode emerging as the most influential predictor of profitability. While the model provides useful directional insights, its explanatory power suggests that key cost-related variables—such as shipping costs, distribution costs, and shipping distance—are missing from the current dataset. As a result, it is recommended that Amazon incorporate additional operational cost data and expand the sample size used for modeling. Given Amazon's scale of approximately 12 million items sold per day, leveraging a larger dataset and more comprehensive cost variables would likely result in a more accurate and actionable profit prediction model.

Data Exploration with Tableau

Click Here to Access the Tableau

Amazon's average profit per item varied heavily between city, state, segment, and subcategory. Home office products were the most profitable items with an average profit of $34.01 on each item. The average profit of cities varied widely, ranging from -$535.87 in Burlington, North Carolina to $748.01 in Lafayette, Indiana. The most profitable state was Vermont with $204.09 while Colorado had a $36.64 average loss. Copiers had by far the most profit with an average profit of $816.01 while tables were sold at an average loss of $55.91.

Statistical Analysis

Introduction to the Data

The raw data we were given for this scenario contained lots of variables. We were given data such as profit per item, revenue, quantity of items in the order, ship mode, segment, city and state, region, and ship and order date. To clean up the data, we modified a few of the variables to better fit our regression model. This included creating dummy variables for a few of the variables, including separating ship mode into 4 subsections: standard class, first class, second class, and same day delivery. We also created a dummy variable and separated the category of products into subsections: furniture, technology, and office supplies. Another dummy variable

[1] ChatGPT (OpenAI) was used to assist with drafting and editing portions of the introduction. All analytical decisions, interpretations, and conclusions are the author's own.

separated region into subcategories: East, South, West, and Central. Finally, a dummy variable was made separating segment into subcategories: consumer, corporate, and home office.

Descriptive Statistics

For the data included in our final regression model, we ran Descriptive Statistics for it in Excel. Not included is the categorical data that was used, such as the dummy variables for shipping mode and for subcategory of product. The shipping class that was most used in the data given was Standard Class, on 5,910 orders. Same Day Delivery was used the least, on only 540 deliveries. The average discount percentage given on the orders was right around 15%. Profit per item averaged around $7.78, ranging from -$1,320 to $1,680. The average quantity of items per order was about 4 items, with the most being 14 items in one order. Finally, the average revenue of each individual item was $230.29.

Attempted Models

In trying to determine the best regression model for predicting profit per item of Amazon orders, we ran 5 different regression models to test out all the data that we were given. The first model that we ran tested 13 different variables, with dummy variables to separate the subcategory of product, as well as shipping mode and region separations. Before running the model, we removed 6 different order ids (1069, 921, 1067, 9, 154, and 1068) as these orders did not have some of the data we needed. This model came back with a 0.2658 R^2. To improve the model, we next set up a dummy variable for separating the quarter of the year that the item had been ordered in. we believed that certain times of the year yielded higher revenue due to the holiday season, which we believed would lead to higher profit. However, the second model's R^2 dropped tremendously down to just 0.03. The next model that we did included a dummy variable for separating the regions, in which we determined again that the regions had no impact and was not benefiting our model anymore. For the fourth model, we created a dummy variable for specific product categories such as furniture, office supplies, and technology, along with the main variables we deemed significant such as discount percentage, quantity, and revenue. This again came back with a 0.2658 R^2, which led us to believe that we had found a best model possible with the data given. We added back the dummy variables for shipping mode for a final regression model, which once again led us to a R^2 of 0.2658. We saw no common violations in the final model either.

Final Regression

Final Population Regression

$$Profit\ Per\ Item = \beta_0 + \beta_1(Discount\%) + \beta_2(Furniture_{(d)}) + \beta_3(Office\ Supplies_{(d)}) + \beta_4(Quantity) + \beta_5(Revenue) + \beta_6(Standard_{(d)}) + \beta_7(First\ Class_{(d)}) + \beta_8(Second\ Class_{(d)}) + \varepsilon$$

Final Estimated Sample Regression

$$\widehat{Profit\ Per\ Item} = 16.68 - 0.58(Discount\%)$$

$$-12.07\ (Furniture_{(d)}) + 0.59(Office\ Supplies_{(d)}) - 2.53(Quantity) + 0.04(Revenue) + 3.12(Standard_{(d)}) + 1.71\left(FirstClass_{(d)}\right) + 0.89(SecondClass_{(d)})$$

R^2 and Standard Error Interpretations

Click here to view the final regression output for the next few sections

$R^2 = 0.2658$: Using this model, we are 26.58% of the way towards perfectly predicting profit per item of Amazon orders.

Standard Error = 48.25: Using this model, predictions of profit per item of Amazon orders are off by an average of $48.25

Significance Testing

We found that 5 variables were statistically significant because their respective p-values were less than 0.10

- Discount %: P-Value is 3.89E-132
- $Furniture_{(d)}$: P-Value is 5.68E-15
- Quantity: P-Value is 1.03E-29
- Revenue: P-Value is 0
- Standard $shipping_{(d)}$: P-Value is 0.007

Coefficient Interpretations

The intercept is at 16.68, this means there is an expected profit per item of $16.68 in the baseline category. The baseline category is technology with both a discount and quantity of 0.

As discount percentage increases by 1 percentage point, profit decreases by $0.58, on average and all else constant.

Profit per item is $12.07 lower for furniture items than for technology items, on average and all else constant.

Profit per item is $0.59 higher with office supplies than technology products, on average and all else constant.

As the quantity of items increases by 1, profit per item decreases by $2.53, on average and all else constant.

As revenue increases by $10, profit per item increases by $0.40, on average and all else constant.

Profit per item is $3.12 higher for standard shipping than same day shipping, on average and all else constant.

Profit per item is $1.71 higher for first class shipping than same day shipping, on average and all else constant.

Profit per item is $0.89 higher for second class shipping than for same day shipping, on average and all else constant.

Common Violation Checks

Heteroskedasticity and Non-Linear Patterns

Click here to view the residual plots

We do not have any evidence of heteroskedasticity and nonlinear patterns. In the residual plots, we did not see any funnel shapes or curved patterns.

Endogeneity and Multicollinearity

Click here to view the correlation matrix for violation checks

Multicollinearity: We do not have any evidence of Multicollinearity.

Endogeneity: All the correlation between the x-variables and the residuals are extremely close to 0. This indicates we do not have issues with endogeneity. This means we can trust the coefficients estimated in the regression.

Relevance of Findings

Important Trends and Relationships

Through our regression model, we found that discount percentage has a strong and statistically significant negative relationship with profit per item. As the discount percentage increases by 1 percentage point, profit per item decreases by $0.58 on average and all else constant. This indicates that as Amazon offers a higher discount on their products, the less profit they make off of it. There was also a positive significant relationship with using standard shipping and profit per item. The profit per item was $3.11 more for standard shipping than for using another method, same day shipping. This was the highest profit increase compared to all the other shipping methods. This indicates to us that standard shipping is likely the best method to use when it comes to generating the most profit per item.

Surprising Findings

We were surprised that there were not a lot of variables that had a large impact when it came to affecting the profit per item. The highest impacting variable was that furniture made $12.07 less profit per item than technology items did, on average and all else constant. We did believe that there would be more variables with a stronger impact on profit per item. Seeing that there wasn't makes us think that more data and more variables need to be tested to find what is the most important when it comes to predicting profit per item. It was also interesting for us to see that the correlation matrix showed there was a perfect negative correlation between the central region and quarter 4 of the year. We were not sure why this could have occurred, but the variables were not statistically significant, so we did not look any further at this time.

Important Non-Significant Variables

We made separate variables, of each of the 4 quarters in the year. Using each of the 4 quarters, we believed quarter 4 would be significant and have an impact on the profit per item. This would be because lots of holidays like Thanksgiving and Christmas where lots of people are buying things fall into that quarter. However, this variable ended up having a non-significant impact in terms of the 10% significance level.

Recommendations

Next Steps

For our next steps, we would like to get some more data to try and create a better regression model. Specifically, we think that data on the behind the scenes stuff; like cost of goods, cost of labor for making the goods, and distance to ship orders. We believe that information on these variables would give us more insight on profit per item and allow us to make a model closer to perfectly predicting this information.

Future Testing

We wish to do further testing into creating a better regression model to get closer to perfectly predicting profit per item at Amazon. We could not get our R^2 as high as we had hoped to, and we believe that by getting some more data and different variables to test, such as the ones listed above and possibly more variables, we could create a better model. We think more data on prices of the items, operating expenses, and the costs to produce the items could get us a closer regression model for predicting profit per item.

Conclusion

Recap of Findings and Next Steps

In summary, this analysis demonstrated that while it is possible to model Amazon's average profit per item using historical order data, the current regression model explains only a limited portion of profit variability. Among the variables analyzed, shipping mode was identified as the most significant driver of profitability, highlighting the importance of fulfillment and logistics decisions in influencing margins. However, the model's modest explanatory power indicates that critical cost-related factors are not captured in the existing dataset. To improve predictive accuracy and strengthen decision-making, future analyses should incorporate detailed shipping, distribution, and fulfillment cost data, as well as measures of delivery distance and operational complexity. Additionally, expanding the dataset to include a larger volume of orders would better reflect Amazon's scale and improve model robustness. These next steps would enable the development of a more precise and actionable profit prediction model, ultimately supporting more informed operational and strategic decisions.

Contact Information

If you have any questions at all, or would like more clarification on any of the information that has been provided, please do not hesitate to reach out to our team at profitabilityteam@amazon.com. We can also be reached via phone for urgent communication at 520-005-1010.[2]

Appendix

Figure 1 - Final Regression Model Descriptive Statistics

Descriptive Statistics

Profit per item		*Discount (%)*		*Revenue*		*Quantity*	
Mean	7.778604775	Mean	15.66491299	Mean	230.2852231	Mean	3.790267098
Standard Error	0.566187715	Standard Error	0.20800399	Standard Error	6.287985046	Standard Error	0.022341543
Median	2.742	Median	20	Median	54.432	Median	3
Mode	3.1104	Mode	0	Mode	12.96	Mode	3
Standard Deviation	56.28942472	Standard Deviation	20.67940476	Standard Deviation	625.1408351	Standard Deviation	2.221158446
Sample Variance	3168.499335	Sample Variance	427.6377812	Sample Variance	390801.0637	Sample Variance	4.933544844
Kurtosis	354.1035931	Kurtosis	2.38601539	Kurtosis	304.8955631	Kurtosis	1.958335434
Skewness	7.616169112	Skewness	1.679713613	Skewness	12.98245092	Skewness	1.271606596
Range	2999.9908	Range	80	Range	22638.036	Range	13
Minimum	-1319.9956	Minimum	0	Minimum	0.444	Minimum	1
Maximum	1679.9952	Maximum	80	Maximum	22638.48	Maximum	14
Sum	76883.7296	Sum	154832	Sum	2276139.145	Sum	37463
Count	9884	Count	9884	Count	9884	Count	9884

Click here to return back to Descriptive Statistics section

[2] ChatGPT (OpenAI) was used to assist with drafting and editing portions of the conclusion. All analytical decisions, interpretations, and conclusions are the author's own.

Figure 2 - Final Regression Model Output

Regression Statistics	
Multiple R	0.515579007
R Square	0.265821712
Adjusted R Square	0.265226935
Standard Error	48.25067842
Observations	9884

ANOVA

	df	SS	MS	F	Significance F
Regression	8	8324015.241	1040501.905	446.926423	0
Residual	9875	22990263.69	2328.127968		
Total	9883	31314278.93			

	Coefficients	Standard Error	t Stat	P-value	Lower 90.0%	Upper 90.0%
Intercept	16.68065743	1.747916955	9.543163581	1.71473E-21	13.80532015	19.55599471
Discount (%)	-0.584811709	0.023534469	-24.84915643	2.8844E-132	-0.623526098	-0.54609732
Furniture (d)	-12.07506137	1.549423941	-7.793258546	7.18861E-15	-14.62387607	-9.526246674
Office Supplies (d)	0.594466847	1.32024126	0.450271375	0.652524661	-1.577340518	2.766274213
Quantity	-2.529401295	0.223425225	-11.32101935	1.56963E-29	-2.896937566	-2.161865024
Revenue	0.042346372	0.000814832	51.96947	0	0.041005967	0.043686777
Standard (d)	3.115688515	1.235412521	2.521982303	0.011685152	1.083425099	5.14795193
First (d)	1.708155446	1.347408949	1.267733488	0.204923032	-0.508342983	3.924653875
Second (d)	0.894501889	1.531264939	0.5841588	0.559126821	-1.624441105	3.413444882

Click here to return back to the sections discussing the regression output

Figure 3 - Final Regression Correlation Matrix

	Profit per item	Discount (%)	Furniture (d)	Office Supplies (d)	Quantity	Revenue	Standard (d)	First (d)	Second (d)
Profit per item	1								
Discount (%)	-0.23245835	1							
Furniture (d)	-0.053573606	0.044072057	1						
Office Supplies (d)	-0.043966378	0.006927475	-0.639412306	1					
Quantity	-0.007736026	0.00892693	0.000201656	0.00528776	1				
Revenue	0.446535901	-0.028457817	0.099889068	-0.218962023	0.200489202	1			
Standard (d)	0.022470856	0.013872336	-0.001632749	-0.005183197	-0.003911499	0.003402394	1		
First (d)	0.008178088	0.018424714	-0.001123619	-0.009173247	-0.01681823	-0.000883588	-0.003441537	1	
Second (d)	-0.010537423	-0.013722936	0.001032555	0.002845485	-0.002684239	-0.007840073	-0.598398751	-0.00318504	1

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Figure 4 - Quantity Variable Residual Plot

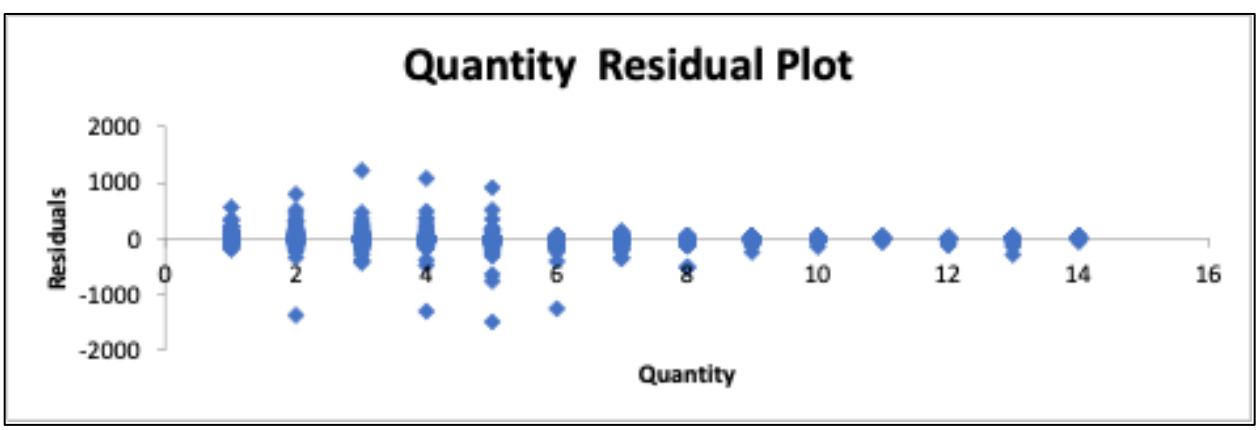


Figure 5 - Discount % Variable Residual Plot

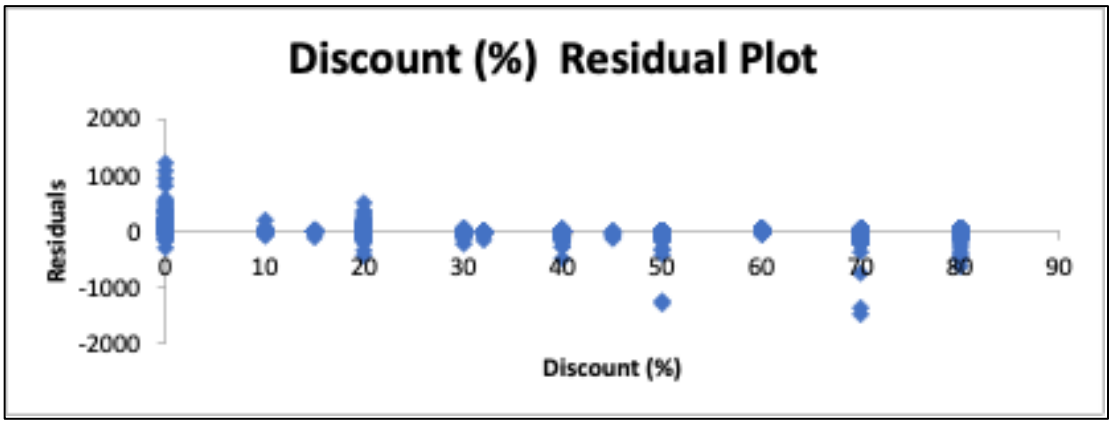
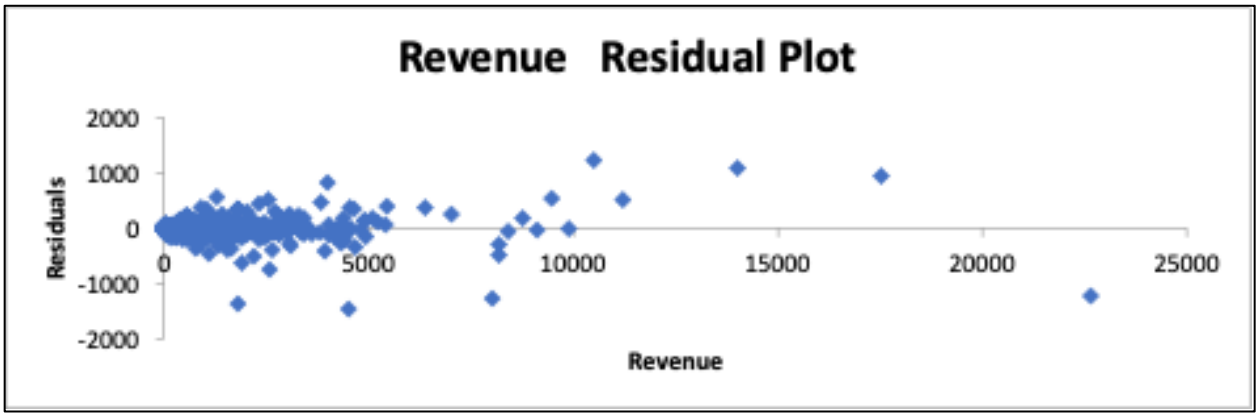


Figure 6 - Revenue Variable Residual Plot



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Sources Consulted

OpenAI. (2025). *ChatGPT* (GPT-5.2) [Large language model]. https://chat.openai.com/